Honeywell

Asset Performance Management

DAS Plugin

User Guide

Release 540.1

August 2023

DISCLAIMER

CONTENTS

CHAPTER

1 ABOUT THIS GUIDE

This document describes how to install and configure the Asset Performance ManagementDAS Plug-in.

1.1 Revision history

Revision	Date	Description
1	06/23	R540.1 release of this document.

1.2 Related documents

The following list identifies publications that may contain information relevant to the information in this document.

Document name	Document number
Honeywell Intuition Data Access Service User's Guide	-
Honeywell Intuition Core System User's Guide	-
Asset Performance Management Configuration Guide	APMDOC-X391-en-R540.1

CHAPTER

2 INTRODUCTION

2.1 Overview

The Asset Performance Management Plug-in connects to your Asset Performance Management model through the Intuition Data Access Service allowing the data to be viewed and used in applications that make use of DAS.

CHAPTER

3 INSTALLING THE ASSET PERFORMANCE MANAGEMENT DAS PLUG-IN

If you are working on a computer with Asset Performance Management installed (such as a single server or the Application Server), the DAS Plug-in is already installed, and you can begin with configuration (see Configuring the Asset Performance Management DAS Plug-in User).

If you are working on a client computer without a local Asset Performance Management installation, or if the plug-in was uninstalled from the server, you need to install the DAS Plug-in.

3.1 Pre-requisites

- Data Access Service (DAS) R255.1, R240, or R230 is installed

3.2 Installation

1. From the installation media, in the `\Installer\Components\SentinelDASPlugin` folder, right-click the `setup.exe` file and select **Run as Administrator**.
2. Click **Next**.

   The License Agreement page appears.
3. Read through the license agreement clauses, select **I accept the terms in the License agreement**, and click **Next**.

   Data Source Plug-in Selection page appears.
4. Click **Next**.

   The **Feature Selection** page appears.
5. Click **Next**.

   The **Summary** page appears.
6. Click **Finish** to complete the installation.

CHAPTER

4 CONFIGURING THE DAS PLUG-IN USER

4.1 Updating Data Source User Permissions

Adding User to Windows Reliability Engineer Group

To ensure the Plug-in can connect to the server, the user who assigned to the data source (on the Application Server) must have Reliability Engineer access in MES on the Server.

It is recommended to add the user to Windows Reliability Engineer group (which is added to the required security role by default).

Perform the following steps to update the data source user permissions for adding user to Windows Reliability Engineer group.

1. **Start > Server Manager**.
2. On the server tree, choose **Configuration > Local Users and Groups > Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the name of the user assigned to the data source, and then click **OK**.
6. Click **OK** to close the **Reliability Engineers Properties** dialog box.

Adding User to the Role Individually

Perform the following steps to update the data source user permissions for adding an individual user.

1. On the Security page, select the **User Assignment** tab.
2. From the **Associated Roles** list, select **Reliability Engineer**.
3. In the **Role Members** area, in the **Windows User or Group Name** box, type the name of the user, and then click **Associate**.

   The user appears in the Associated Members list.

CHAPTER

5 CONNECTING TO ASSET PERFORMANCE MANAGEMENT

5.1 Overview

To connect Asset Performance Management system to Intuition Executive, you must create a data access service instance for Asset Performance Management.

5.2 Creating the Data Access Instance Installation Files

To create the data access service instance, you must first create the installation files using the Service Generator tool. A complete installation setup or Bill of Material for the instance is created.

Run the Service Generator tool on the Intuition Application Server and the Intuition Web Server, using the same instance name on both.

To create the data access instance

1. Browse to **Start > All Programs > Honeywell > Service Generator**.

   The **Honeywell Intuition - Data Access Service Instance Management Tool** window is displayed.
2. Click the **Generate Service Installer** tab.
3. In the **Service Instance Name** box, type a name for the instance. For example, CommonAccess.

   > **NOTE**
   > Ensure that the instance name does not contain special characters and numbers.

4. The **Connection Template File Location** box displays the location of the template file for storing the created instances. By default, the path for the new files is

   `<Installed drive>:\<installpath>\ProgramData\Honeywell\MES\DataFiles\ConnectionTemplates`

   To change the template location, click the ellipsis button, and then browse to the location.
5. Click **Generate Installer**.

   A BOM (Bill of Material) folder for the instance is created in the path mentioned in step 4.

5.3 Installing the Data Access Service Instance

5.3.1 Overview

The data access instance must be installed on the Intuition Application Server, and then on the Intuition Web Server.

5.3.2 Instance Installation

5.3.2.1 To install the data access service instance

1. Perform one of the following steps:
    a. If the Service Generator is still open from generating the installer, click **Run Installer**.
    b. In the BOM folder, right-click the `setup.exe` file, and then select **Run as Administrator**.
2. Click **Next**.

    The **License Agreement** page appears.
3. Read through the license agreement clauses, select **I accept the terms in the License agreement**, and click **Next**.

    The Honeywell Intuition Application Selection page appears.
4. Select the **Data Access Service - <Instance Name>** component from the list (for example, Data Access Service - CommonAccess), and click Next.

    The **Deployment Features Selection** page appears.
5. Click **Next**.

    The **Feature and Options selection** page appears.
6. Click **Next**.

    The **Application Pool Configuration** page appears.
7. Perform the following steps:
    a. Click **Existing**.
    b. From the **Select Application Pool** drop-down list, select **MESCommon4.0**.
    c. Click **Next**.

    The **Summary** page appears.
8. Click **Install** to complete the installation.

CHAPTER

6 CONFIGURING THE ASSET PERFORMANCE MANAGEMENT PLUG-IN

6.1 Overview

Once the Data Access Service instance is created and installed, you must configure the plug-in to connect with your Asset Performance Management system.

To configure the Asset Performance Management Plug-in, you must complete the following tasks:

1. Add the Plug-in Data Source to the data access instance.
2. Configure Security and Cache Settings.
3. Update the Advanced Configuration Settings (Optional).
4. Enable/Disable UOM (Optional).
5. Update MES Security Settings.

> **NOTE**
> After completing the installation and configuration of the data source, perform an **IISRESET** on both the Intuition Application Server and the Intuition Web Server.

6.2 Add the Asset Performance Management Plug-in Data Source

To add the Asset Performance Management plug-in data source, follow the below steps:

1. On the home page of Intuition, choose **Configuration > Data Access**.
2. In the Data Access configuration window, click the **Services** tab. The Instance name (for example, **CommonAccess**) is displayed.
3. Select the instance name, and then click the plus ⊞ icon to add the Asset Performance Management plug-in data source.

   The **Data Source Configuration** page appears.
4. In the **Name** box, type a name for the connection string to access the Asset Performance Management data source.
5. In the **Description** box, type the description of the connection string.
6. From the Select Plug-in drop-down list, select **AMDataSource**.
7. Click **Select** to add a **Connection String**.

   The External Connection Configuration dialog box appears. You can create a new connection string or select an existing connection string.

   Example of connection string:

   ```
   https://<WebServerName>/MES/AM/
   ```

   Where the **<WebServerName>** is fully qualified domain name of the Asset Performance Management Server.

8. Follow the below steps to select an existing connection sting:
    a. Click on existing connection string.

       The **Edit Connection String** section appears.
    b. Enter the description of connection string.
    c. Click **Test Connection**.
    d. Click **Save**.
    e. Click **Apply**.

9. Follow the below steps to create a new connection string:
    a. Click plus ⊞ icon.

       The **Create Connection String** section appears.
    b. In the **Connection String Name** box, type a name for the connection string.

       Example:

       AssetManagerServicesPrefix.
    c. In the Connection String box, type the connection string for the data source plug-in to connect to the Asset Performance Management data source. This is the wcf endpoint for your Asset Performance Management server.

       Example:

       Enter `https://<WebServerName>/MES/AM/` in the connecting string box, where the **<WebServerName>** is fully qualified domain name of the Asset Performance Management Server.
    d. In the Description box, type a description for the connection string.
    e. Click **Test Connection**.
    f. Click **Save**.
    g. Select the new connection sting and click **Apply**.

> **NOTE**
> The Test Connection feature may not succeed when connection to a Asset Performance Management data source. To check the configuration settings, complete the configuration and security settings, and look for the Asset Performance Management data in the Intuition model browser after performing the IISRESET on the Intuition Application Server and Intuition Web Server.

10. In the **Cache Manager Plugin** section, select **Enabled** to read the cached data.
11. Clear Get All Rows.
12. Select **Automatic Sync** if you want to refresh the cached data and automatically synchronize it with the Asset Performance Management data once the cached data is expired.
13. In the **Data Retention** box, type the time required for the data to be retained in cache before it is expired.
14. Click **Create**.

    The data source is displayed in the Data Access configuration window.

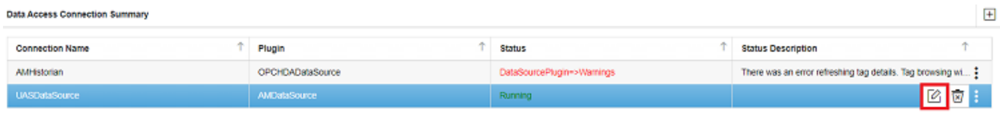
6.3 Advanced Configuration Settings

The following additional configuration settings scan be configured to match your system requirements.

To update the advanced connection settings, follow the below steps:

1. Click three dot against the data source to edit.
2. Click **Edit** icon.

   The **Edit Connection** page appears.

3. Click **Advance Settings** under **Data Source Plugin** to expand the list of settings.
4. Update the following fields based on your system requirements:
   a. **Backup and Restore**: Set the value to **True** to enable backup and restore feature in AllRows Cache.
   b. **Backup Folder**: The backup and restore location of the files:
   c. **Maximum Cached Entities**: The maximum number of entities the system should hold in cache.
   d. **Time Series Cache Mode**: The different modes are **Predictive**, **Incremental** and **Proactive**.
   e. **Time Series Tag Retention**: The time tag cache will remain active after its last access.
   f. **Time Series Value Retention**: Time a value will remain cached after its last access.
   g. **Time Series Minimum Refresh Rate**: The minimum time delay between the checking if new time series data.
   h. **Time Series Maximum Refresh Rate**: The maximum time delay between the checking if new time series data.
   i. **Maximum Tags Cached**: Maximum tags that can be cached at a time.
   j. **Maximum Time Series Values Cached**: The maximum time series values that can be cached at a time.
   k. **Maximum Age Of Time Series Data**: The maximum age of any value before it will be refreshed.

5. Click **Advance Settings** under **Data Source Plugin** to expand the list of settings.
6. Update the following fields based on your system requirements:
   a. **Asset log history days**: The number of days in the past from which to retrieve asset logs. The default is 100 days.
   b. **Event history days**: The number of days in the past from which to retrieve events. The default is 100 days.
   c. **Load children events**: Whether or not to assets show events of child assets. The default is **true**.
   d. **Fault comment history days**: The number of days in the past from which to retrieve fault comments. The default is 100 days.
   e. **Tag source name**: The name of the tag source. The default is **asmgr**.
   f. **Tag Cache Browsing Enabled**: By default the value is set to **false** and updating the value to **true** will enable you to view the UOM at instance level.
   g. **Tag Sync Period**: Time period between each tag synchronization.
   h. **Aggregation List**: Comma separated aggregations supported by the plugin. Below table contains all the supported aggregations:

Aggregation	Description
AVERAGE	Returns the average of the retrieved raw data.
COUNT	Returns the count of the retrieved raw data.

Aggregation	Description
INTERPOLATIVE	Returns the interpolated data for the mapped historical tag.
MAXIMUM	Returns the maximum of the retrieved raw data.
MINIMUM	Returns the minimum of the retrieved raw data.
RANGE	Returns the range of the retrieved raw data.
REGDEV	Returns the standard deviation of the retrieved raw data.
STDEV	Returns the standard deviation of the retrieved raw data.
TOTAL	Returns the total of the retrieved raw data.
VARIANCE	Returns the variance of the retrieved raw data.
RAWTREND	This aggregate picks raw data point for each sample duration over a period of time reducing the number of data points. This results in an efficient trend plot. Example: If you want to view raw data trend for a duration of 1 month. Assume Raw data is present after every 15 seconds, so one month will have (30*24*60*4=172800) data points. But for any UI Trend 2000 points are sufficient. Now the Raw Trend aggregate will return only 2000 points which are spread across selected time range. If the data is missing for a sample, the Raw Trend returns NAN.

i. **Tag Cache Limit**: Maximum size of tag cache for browsing/searching.
j. **Instance UOM**: Enable the UOM checkbox to view the UOM at asset instance level.
k. **Bindings**: The entity bindings.

> **NOTE**
> To return results for the Event History, Fault Comment History, or Asset Logs, Asset Performance Management requires start and end times. However, the Intuition Model Tree does not provide or require a time range. To ensure these results display as expected, set the number of days to display in the **Advanced Settings: Event history days, Fault comment history days,** and **Asset log history days**, as described above.

7. Click **Save**.

6.4 Enable/Disbale Instance UOM

This setting is used to view the instance level unit through Asset Performance Management DAS plugin

Follow the below steps to enable/disable instance UOM

1. Follow steps 1 to 3 of Advanced Configuration Settings.
2. Ensure **Tag Cache Browsing Enabled** value is set to **True**.
   This will enable you to view the UOM at asset instance level.
3. Select **Instance UOM** check box to enable it.
4. Perform **IISRESET**.
5. Navigate to the following folder
   <InstallLocation>\Program Files (x86)\Honeywell\MES\DataFiles\TagCaches**SentinelDASInstance_{Connection Name}**
6. Open .XML file and validate whether instance level UOM is generated in the tagcache file.

> **ATTENTION**
> If the instance level UOM is not generated, delete the existing TagCache file and perform an IISRESET.

6.5 Update MES Security Settings

To ensure the Asset Performance Management Plug-in can connect to the server, permission to access the data source needs to be added.

On the MES Security page, the data source service name (for example, CommonAccess) appears under the Data Access Configuration node in the **Tasks** list.

To update the Intuition Services user permissions, follow the below steps:

1. On the **Intuition Security** page, select the **Roles** tab.
2. Perform the following steps:
   a. From the Roles list, select **Data Access Services**.
   b. On the **Tasks** list, under **Data Access Configuration**, select the data source instance name. For example, **CommonAccess**.
   c. Click **Save**.
3. Perform the following steps:
   a. From the Roles list, select **Reliability Engineer**.
   b. On the **Tasks** list, under **Data Access Configuration**, select the data source instance name. For example, **CommonAccess**.
   c. Click **Save**.

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